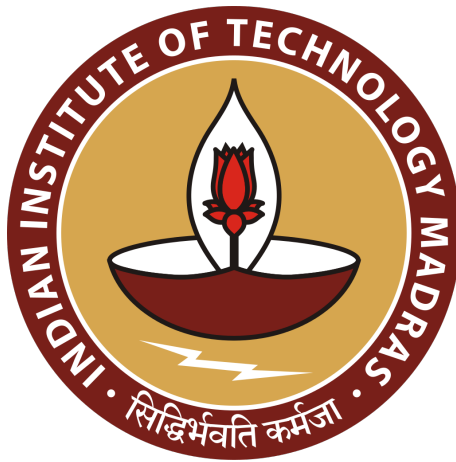


Lab Assignment - 3

Analysing Protein Databases



BT3040 - Bioinformatics

Even Semester 2026

Course Instructor: Professor Michael Gromiha

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Department of Biotechnology

Indian Institute of Technology, Madras

Question 1

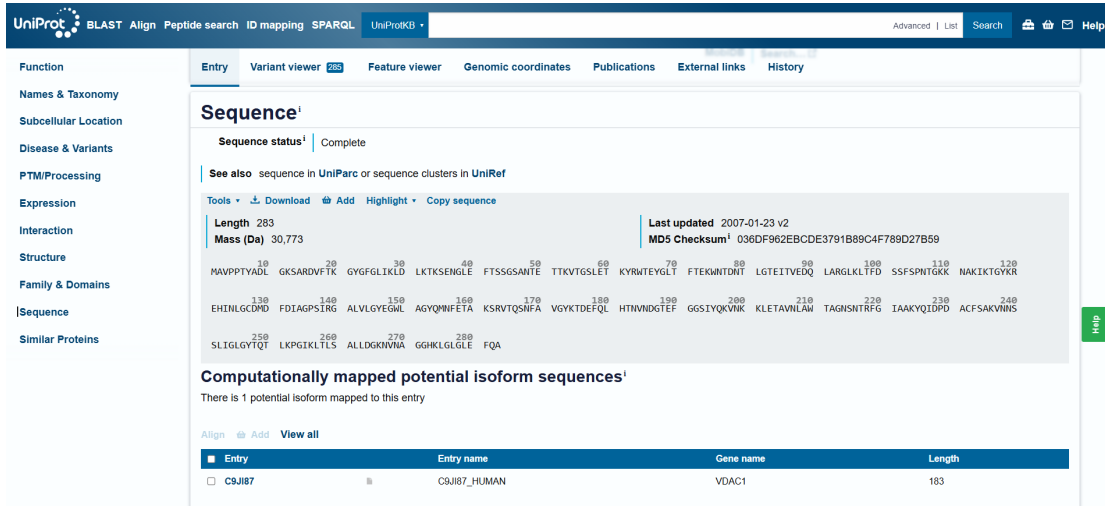


Figure 1.1: Amino Acid sequence of *human mitochondrial β barrel membrane protein VDAC1*

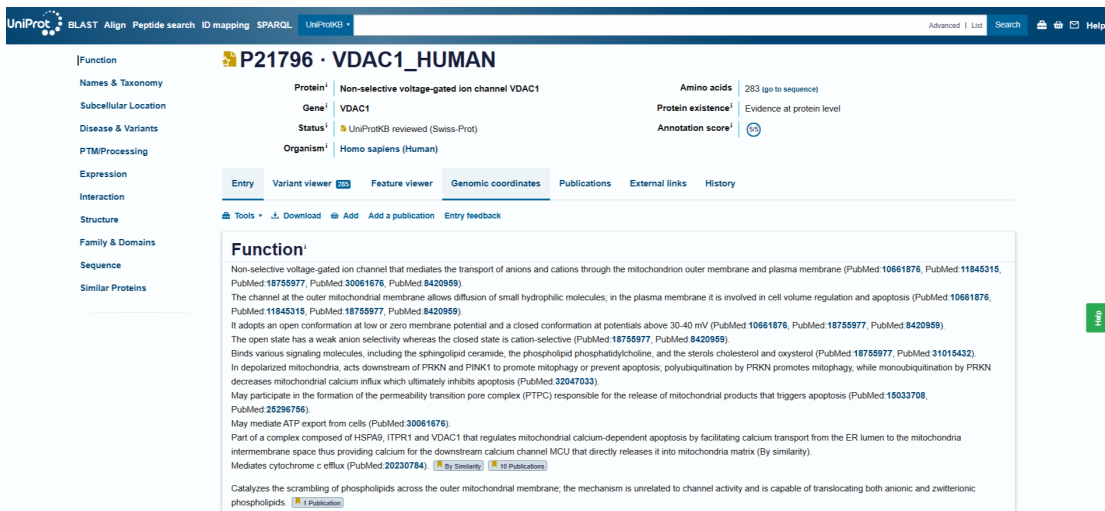


Figure 1.2: Function of *human mitochondrial β barrel membrane protein VDAC1*

The protein contains **19 transmembrane segments**, which are shown below along with their respective positions within the amino-acid sequence.

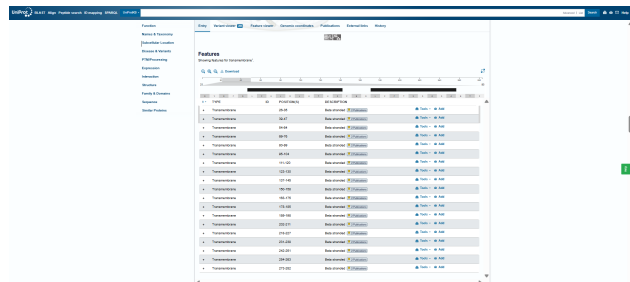
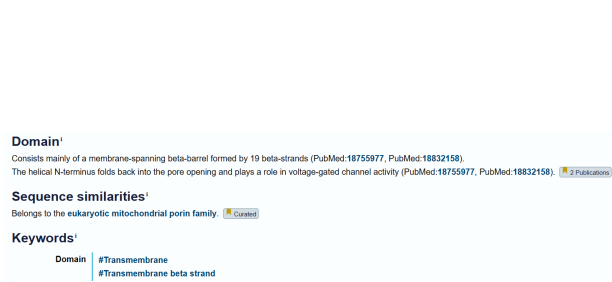
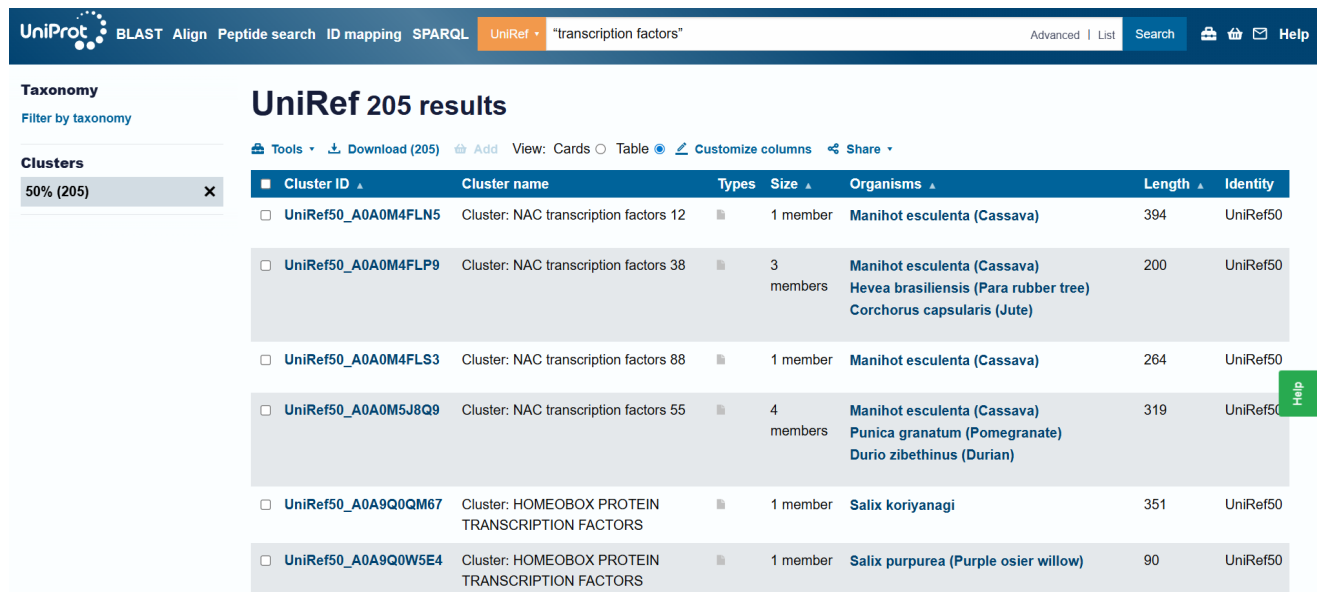


Figure 1.3: Transmembrane Domains

Figure 1.4: Domain Locations

Question 2



UniProt BLAST Align Peptide search ID mapping SPARQL UniRef "transcription factors" Advanced | List Search

Taxonomy
Filter by taxonomy

Clusters
50% (205) x

UniRef 205 results

Tools Download (205) Add View: Cards Table Customize columns Share

Cluster ID	Cluster name	Types	Size	Organisms	Length	Identity
UniRef50_A0A0M4FLN5	Cluster: NAC transcription factors 12		1 member	Manihot esculenta (Cassava)	394	UniRef50
UniRef50_A0A0M4FLP9	Cluster: NAC transcription factors 38		3 members	Manihot esculenta (Cassava) Hevea brasiliensis (Para rubber tree) Corchorus capsularis (Jute)	200	UniRef50
UniRef50_A0A0M4FLS3	Cluster: NAC transcription factors 88		1 member	Manihot esculenta (Cassava)	264	UniRef50
UniRef50_A0A0M5J8Q9	Cluster: NAC transcription factors 55		4 members	Manihot esculenta (Cassava) Punica granatum (Pomegranate) Durio zibethinus (Durian)	319	UniRef50
UniRef50_A0A9Q0QM67	Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS		1 member	Salix koriyanagi	351	UniRef50
UniRef50_A0A9Q0W5E4	Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS		1 member	Salix purpurea (Purple osier willow)	90	UniRef50

Figure 2.1: UniRef "transcription factors" clustered based on sequence identity (50%).

The code to determine the total number of clusters and sequences from the FASTA file is given below (The FASTA File itself is added to the attached google drive):

```
# Question 2

import pprint
import re

fasta_file = "/content/drive/MyDrive/BT3040-Assignments/Lab Assignment - 3/
uniref_transcription_factors_fasta_file.fasta"
with open(fasta_file_path, "r") as f:
    lines = f.readlines()

seq_count = 0
cluster_count = 0
for line in lines:
    if line[0] == ">":
        match = re.search(r"n=(\d+)", line)
        number = int(match.group(1))
        seq_count+=number
        cluster_count+=1

print("The number of sequences:", seq_count)
print("The number of clusters:", cluster_count)
```

The output generated is shown below

```
The number of sequences: 2748
The number of clusters: 205
```

There are 205 clusters (or representative sequences) and 2748 sequences (within the clusters) of transcription factors with 50% sequence identity.

Question 3

Among all the protein sequences from *Homo sapiens*, there are

- 51,801 clusters with 50% identity with 51,801 representative sequences.
- 103,017 clusters with 90% identity with 103,017 representative sequences.
- 240,363 clusters with 100% identity with 240,363 representative sequences.

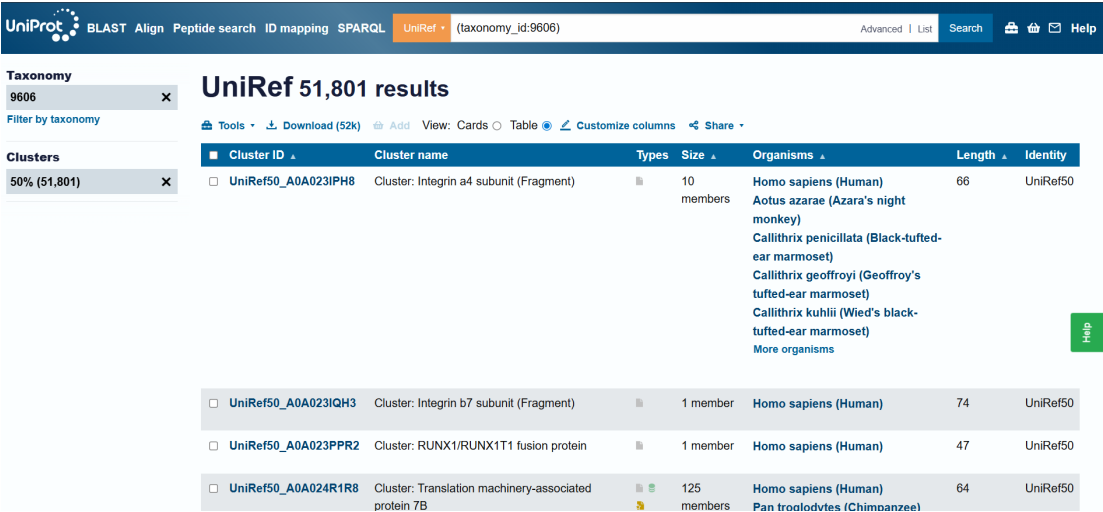


Figure 3.1: protein sequences from *Homo sapiens* with 50% sequence identity

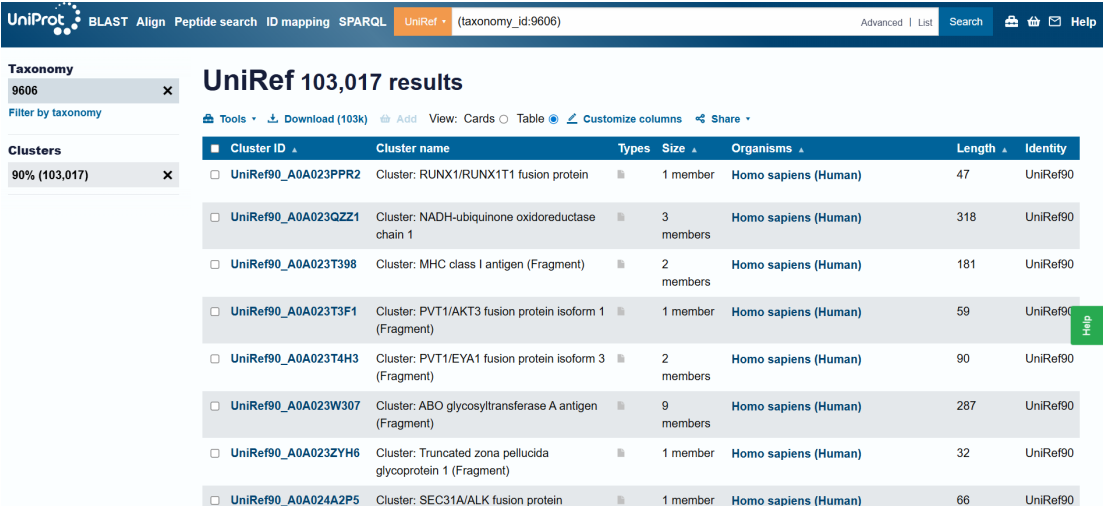


Figure 3.2: protein sequences from *Homo sapiens* with 90% sequence identity

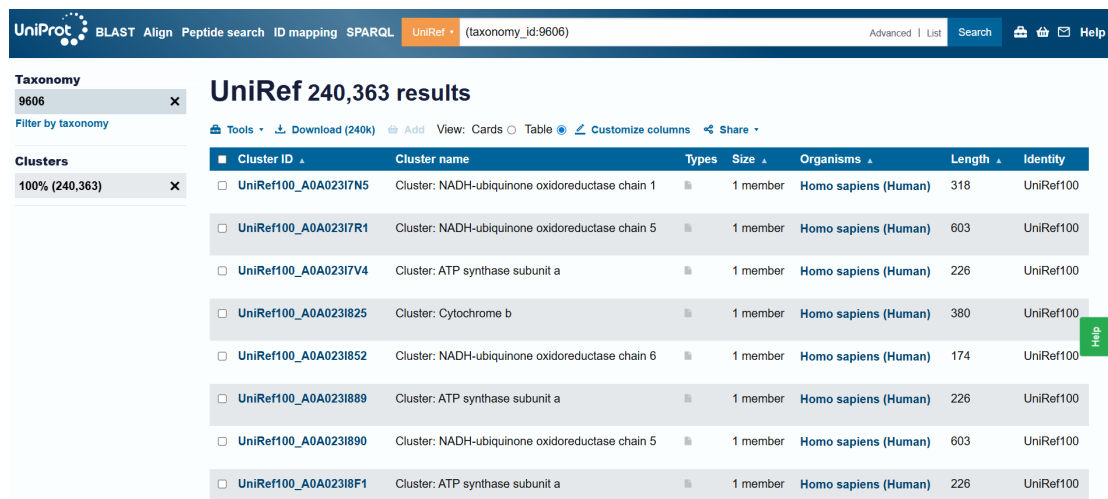


Figure 3.3: protein sequences from *Homo sapiens* with 100% sequence identity

Question 4

In UniProt, 17,252 *Mus musculus* protein sequences are manually annotated/reviewed. Of these manually annotated sequences, 2,619 are associated with PDB (3D Structure).

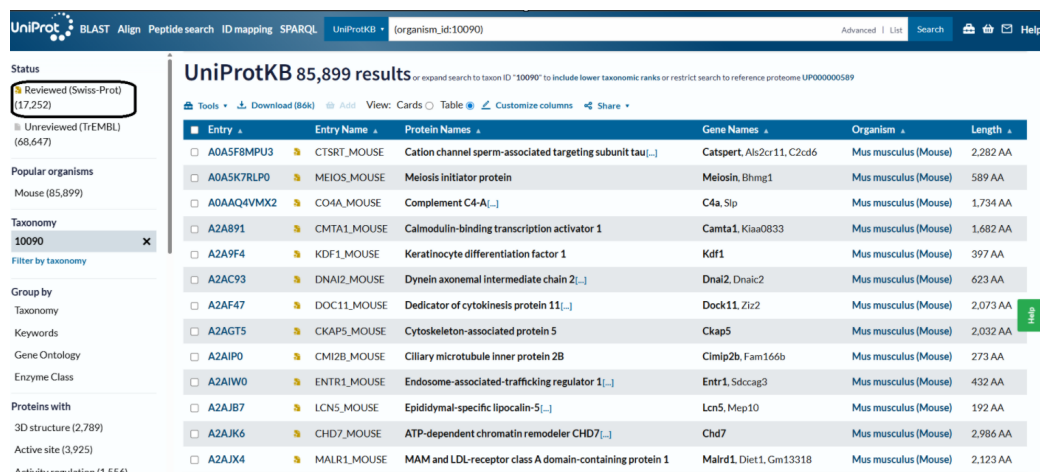


Figure 4.1: *Mus musculus* protein sequences that have been reviewed

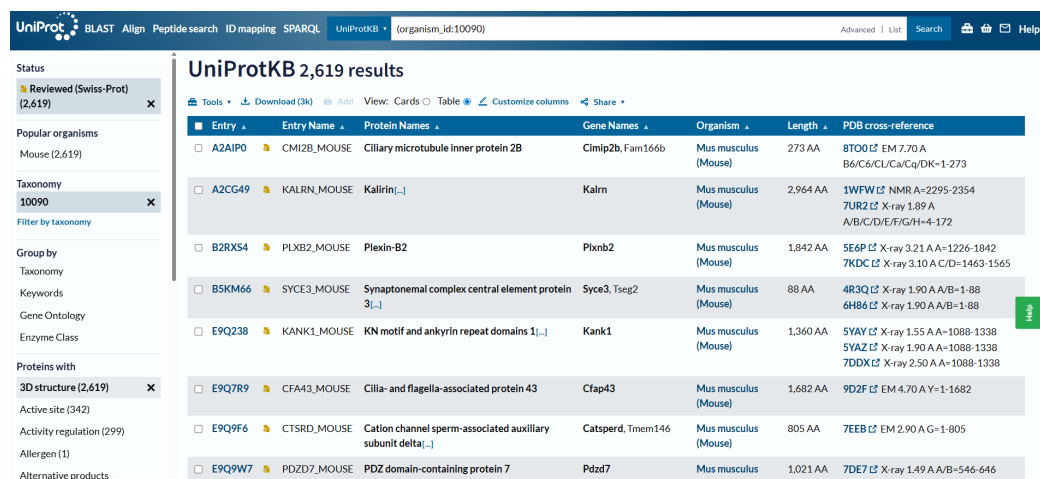


Figure 4.2: *Mus musculus* protein sequences in UniRef associated with PDB 3D structures

Question 5

Status

Reviewed (Swiss-Prot)
(2,619)

Popular organisms

Mouse (2,619)

Taxonomy

10090

Filter by taxonomy

Group by

Taxonomy

Keywords

Gene Ontology

Enzyme Class

Proteins with

3D structure (2,619)

Active site (342)

Activity regulation (299)

Allergen (1)

Alternative products (isoforms) (782)

More items

Protein existence

Protein level (2,619)

UniProtKB 2,619 results

Tools Download (3k) Add View: Cards Table Customize columns Share

Entry	Entry Name	PDB cross-reference	Protein Names
A2AIP0	CM12B_MOUSE	8TO0 EM 7.70 A B6/C6/CL/Ca/Cq/DK=1-273	Ciliary microtubule inner protein 2B
A2CG49	KALRN_MOUSE	1WFW NMR A=2295-2354 7UR2 X-ray 1.89 A A/B/C/D/E/F/G/H=4-172	Kalirin
B2RXS4	PLXB2_MOUSE	5E6P X-ray 3.21 A A=1226-1842 7KDC X-ray 3.10 A C/D=1463-1565	Plexin-B2
B5KM66	SYCE3_MOUSE	4R3Q X-ray 1.90 A A/B=1-88 6H86 X-ray 1.90 A A/B=1-88	Synaptonemal complex central element protein 3
E9Q238	KANK1_MOUSE	5YAY X-ray 1.55 A A=1088-1338 5YAZ X-ray 1.90 A A=1088-1338 7DDX X-ray 2.50 A A=1088-1338	KN motif and ankyrin repeat domains 1
E9Q7R9	CFA43_MOUSE	9D2F EM 4.70 A Y=1-1682	Cilia- and flagella-associated protein 43
E9Q9F6	CTSRD_MOUSE	7EEB EM 2.90 A G=1-805	Cation channel sperm-associated auxiliary subunit delta
E9Q9W7	PDZD7_MOUSE	7DE7 X-ray 1.49 A A/B=546-646 7WEG X-ray 2.00 A A/B=842-946	PDZ domain-containing protein 7
J3QMY9	TO6BL_MOUSE	7QWV X-ray 2.26 A B=559-576	Type 2 DNA topoisomerase 6 subunit B-like
O08580	ERR1_MOUSE	8CEF X-ray 2.49 A C/D/G/H/I/K=50-170	Steroid hormone receptor ERR1
O08582	GTPB1_MOUSE	1KJ2 X-ray 2.71 A P/Q=246-253 1KJ3 X-ray 2.30 A P/Q=246-253	GTP-binding protein 1
O08762	NETR_MOUSE	6H8M X-ray 1.70 A A/B=383-494	Neurotrophin
O08807	PRDX4_MOUSE	3VWU X-ray 3.30 A A/B/C/D/E/F/G/H/I/J=41-274	Peroxiredoxin-4

Figure 5.1: UniProtKB IDs, for manually annotated, PDB associated *Mus musculus* protein sequences

1. The Entry IDs of the top 10 sequences were downloaded in the form of a text file.

2. The text file is then uploaded into the ID mapping tool, and the from and to databases are chosen.

3. The results are displayed below (9 sequences out of 10 were mapped).

ID mapping 9 results found for UniProtKB_AC-ID → STRING

Overview Input Parameters API Request

Tools Download (9) View: Cards Table Resubmit

9 IDs mapped to 9 results
1 ID not mapped:
A2AIP0

From	To
A2CG49	10090.ENSMUSP00000110611
B2RXS4	10090.ENSMUSP00000104955
B5KM66	10090.ENSMUSP00000131766
E9Q238	10090.ENSMUSP00000042177
E9Q7R9	10090.ENSMUSP00000125007
E9Q9F6	10090.ENSMUSP00000108603
E9Q9W7	10090.ENSMUSP00000133273
J3QMY9	10090.ENSMUSP00000136515
O08580	10090.ENSMUSP00000025906

Figure 5.2: ID Mapping UniProtKB IDs to STRING IDs

The sequence A2AIP0 could not be matched, possibly because its protein–protein interactions have not been fully characterized, the protein does not directly interact with other proteins, or sufficient studies have not yet been conducted to properly link the corresponding IDs.

Question 6

The Sequence Size distribution of proteins shown below appears to follow a Poisson distribution with a mode sequence length of ≈ 300 Amino Acids and a mean sequence length of ≈ 400 Amino Acids.

Sequence size

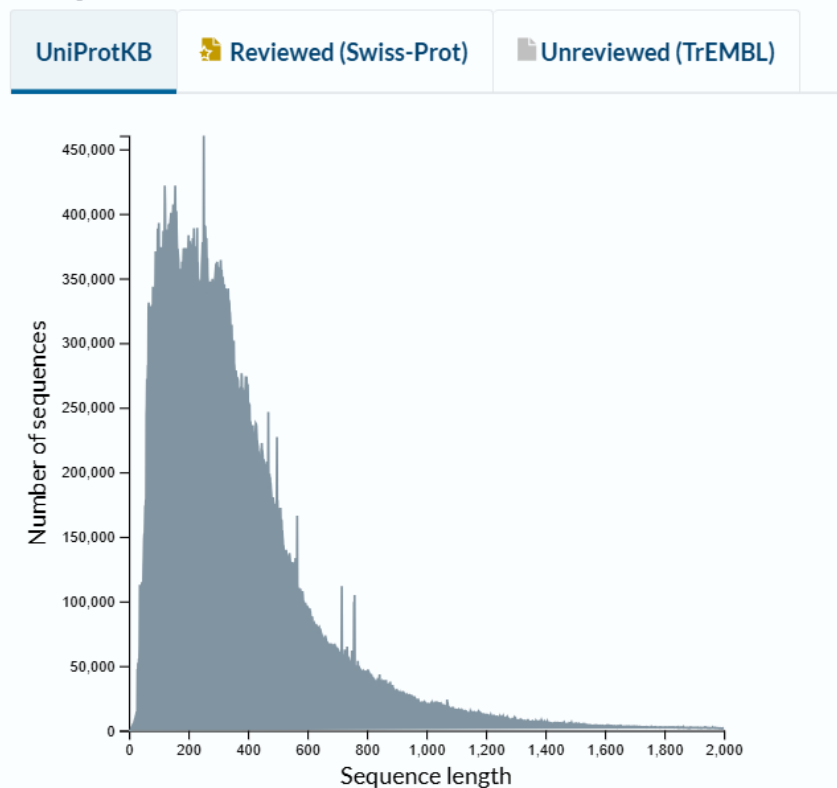


Figure 6.1: UniProtKB database's protein sequence length distribution

The screenshot shows the UniProtKB search results page for the query `(reviewed:true) AND (fragment:false) AND (length:[1 TO 50])`. The results are sorted by sequence length, showing the shortest proteins. The table includes columns for Entry, Entry Name, PDB cross-reference, Protein Names, Sequence, and Length. The results are as follows:

Entry	Entry Name	PDB cross-reference	Protein Names	Sequence	Length
☐ P0DPR3	TRDD1_HUMAN		T cell receptor delta diversity 1	EI	2 AA
☐ P83570	GWA_SEPOF		Neuropeptide GWA	GW	2 AA
☐ P62968	TRH_PIG		Thyrotropin-releasing hormone[...]	QHP	3 AA
☐ P62969	TRH_SHEEP		Thyrotropin-releasing hormone[...]	QHP	3 AA
☐ P62970	TRH_BOMOR		Thyrotropin-releasing hormone[...]	QHP	3 AA
☐ P62971	TRH_NOTVI		Thyrotropin-releasing hormone[...]	QHP	3 AA
☐ P84761	ACL_MACGN		Angiotensin-1-converting enzyme inhibitory peptide	GEP	3 AA
☐ P01858	TUFT_HUMAN		Phagocytosis-stimulating peptide[...]	TKPR	4 AA

Figure 6.2: Proteins with the shortest amino acid sequences

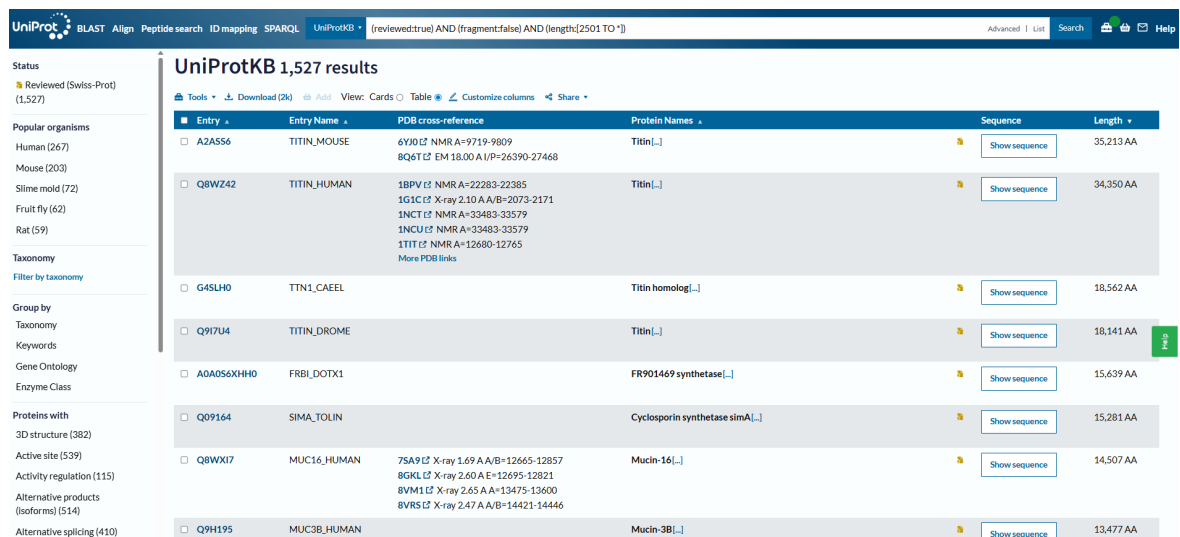


Figure 6.3: Proteins with the longest amino acid sequences

- The shortest sequences are TRDD1_HUMAN and GWA_SEPOF: **2 amino acids**
- The longest sequence is TITIN_MOUSE: **35,213 amino acids**

Amino acid composition

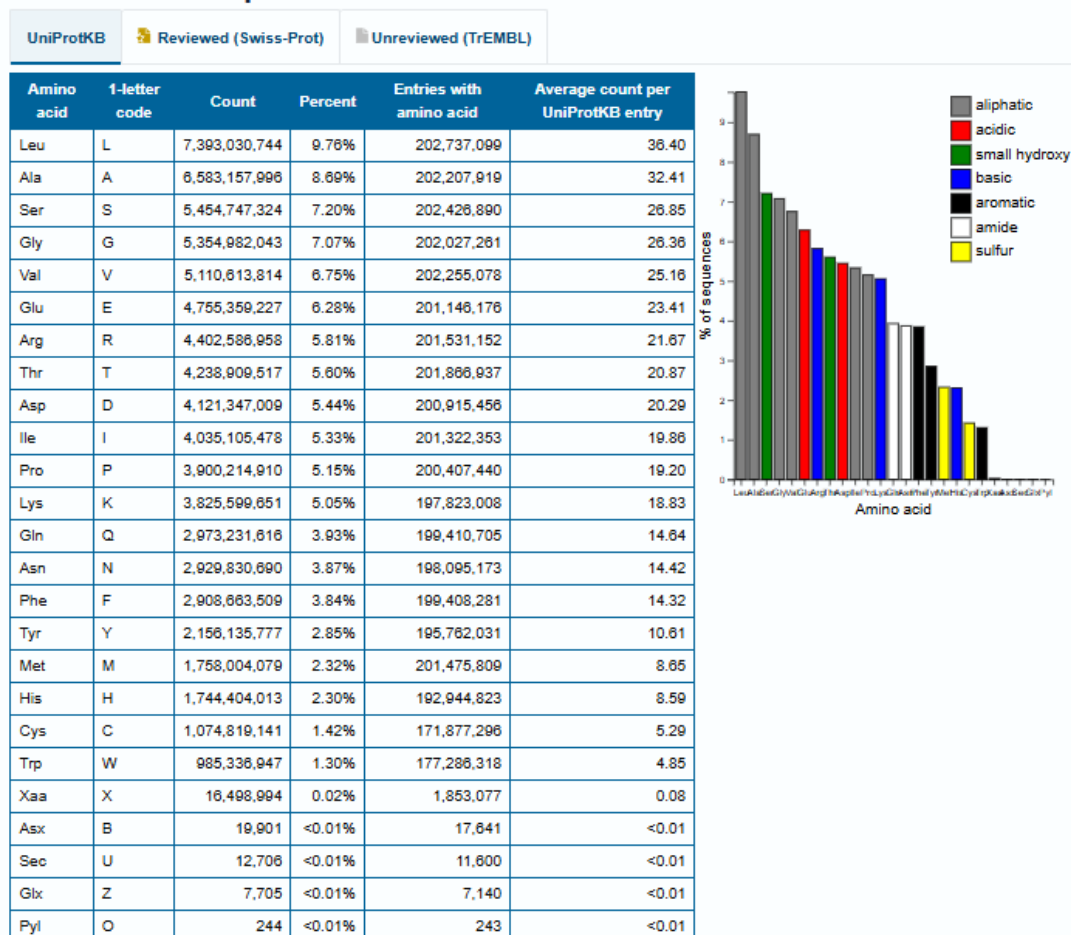


Figure 6.4: Amino Acid occurrence frequency in proteins in UniProtKB

As discussed in class, small non-polar amino acids occur at high frequency, whereas structurally influential amino acids such as cysteine and tryptophan occur at much lower frequencies in proteins.

Amino acid	1-letter code	Count	Percent	Entries with amino acid	Average count per UniProtKB entry
Leu	L	7,393,030,744	9.76%	202,737,099	36.40
Ala	A	6,583,157,986	8.69%	202,207,919	32.41
Ser	S	5,454,747,324	7.20%	202,426,890	26.85
Gly	G	5,354,982,043	7.07%	202,027,261	26.36
Val	V	5,110,613,814	6.75%	202,255,078	25.16
Glu	E	4,755,359,227	6.28%	201,146,176	23.41
Arg	R	4,402,586,958	5.81%	201,531,152	21.67
Thr	T	4,238,909,517	5.60%	201,866,937	20.87
Asp	D	4,121,347,009	5.44%	200,915,456	20.29
Ile	I	4,035,105,478	5.33%	201,322,353	19.86
Pro	P	3,900,214,910	5.15%	200,407,440	19.20
Lys	K	3,825,599,651	5.05%	197,823,008	18.83
Gln	Q	2,973,231,616	3.93%	199,410,705	14.64
Asn	N	2,928,830,690	3.87%	198,095,173	14.42
Phe	F	2,908,663,509	3.84%	199,408,281	14.32
Tyr	Y	2,156,135,777	2.85%	195,762,031	10.61
Met	M	1,758,004,079	2.32%	201,475,809	8.65
His	H	1,744,404,013	2.30%	192,944,823	8.59
Cys	C	1,074,819,141	1.42%	171,877,296	5.29
Trp	W	985,336,947	1.30%	177,286,318	4.85

Link to the google drive containing the python code used in this assignment: https://drive.google.com/drive/folders/12Uhc2XZirFkji6JSEcqnR8sLLXdxpS4_?usp=sharing